Chandra Keshwar Jaiswal

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Summary

Final-year B.Tech CSE student and DRDO Research Intern with hands-on experience building production-style, event-driven backend systems, multi-agent AI workflows, and RAG pipelines. Published researcher (IEEE, Springer) in post-quantum cryptography. Strong foundation in Data Structures & Algorithms, OOP, distributed systems concepts, REST API design, and modular software architecture.

Education

Allenhouse Institute of Technology — Kanpur, India
Bachelor of Technology in Computer Science Engineering; CGPA: 8.72/10 — *Aug 2023 – Jul 2027*

Experience

Defence Research and Development Organisation (DRDO) — Remote
Research Intern – Lattice-Based Post-Quantum Cryptography — *Oct 2025 – Mar 2026*

– Researched lattice-based post-quantum cryptography and secure authentication protocols, contributing to 2 peer-reviewed publications (IEEE SCEECS 2025, Springer ICMLCS 2025).

– Performed formal security validation and mathematical modeling of cryptographic protocols; collaborated with senior researchers on protocol design and technical documentation.

Projects

Ops Agent Hub – Multi-Agent IT Helpdesk Platform | *Python, LangGraph, ChromaDB, MCP, RAG* | GitHub

– Owned the AI layer (1 of 3 engineers) of a **multi-agent LLM/RAG IT helpdesk platform**; refactored a monolithic RAG pipeline into a modular Knowledge Service and migrated **FAISS to ChromaDB vector database** for persistent, incrementally-updatable **semantic search and retrieval**.

– Designed and implemented 4 **LangGraph AI agents** (Triage, Knowledge, Action, Validation) plus a Knowledge-Update agent for re-embedding human-verified resolutions, enabling continuous **RAG knowledge updates** and **human-in-the-loop** escalation with SQLite-backed checkpointing.

– Integrated **Model Context Protocol (MCP)** for tool-based retrieval and workflow automation, including Slack notifications and workflow resumption, separating LLM reasoning from external tool execution.

AI-Powered PR Review Bot | *Python, FastAPI, Redis, ARQ, MongoDB, Tree-sitter, AsyncIO* | GitHub

– Architected an **event-driven AI code-review system** using Python and FastAPI that ingests GitHub PR webhooks, verifies HMAC signatures, and enforces Redis-backed **idempotency** before dispatching retry-safe background jobs through ARQ/AsyncIO.

– Built an **AST-based code analysis and chunking pipeline** using Tree-sitter with sliding-window fallback, and implemented a provider-agnostic **LLM abstraction** supporting Claude, GPT, and Gemini with concurrent inference via `asyncio.Semaphore`.

– Reduced redundant LLM inference using **Redis chunk-level caching**, with schema and line-range validation before posting inline GitHub PR comments; added structured JSON logging, trace IDs, and a React + TypeScript dashboard for **cache hit-rate, latency, and token-cost observability**.

Modular RAG Knowledge Assistant | *React, Python, FastAPI, LangChain, FAISS, Sentence Transformers*

– Built a full-stack **Retrieval-Augmented Generation (RAG)** system for PDF question-answering, implementing document ingestion, text chunking, **embedding generation**, vector indexing, semantic retrieval, and LLM-based response generation through a React frontend and FastAPI backend.

– Integrated **FAISS vector search** with Groq-hosted LLMs through LangChain to generate grounded responses with retrieved context, exposed through **REST APIs** for end-to-end document querying.

CodeSync.ai – AI-Powered Collaborative Development Platform | *MERN Stack, Socket.IO, React* | Live Demo

- Built and deployed a real-time **collaborative coding platform** using the MERN stack, implementing authentication, persistent file-tree management, and multi-user live code editing synchronized through Socket.IO.
- Designed **low-latency real-time state synchronization** to maintain consistency across concurrent editors and sessions; **Winner, Government IDE Bootcamp 2026 (Edition 3, Phase II)**.

Research Publications

- **Challenge-Response Protocol Using Multivariate Polynomial Cryptography** – IEEE SCEECS 2025 [DOI]
- **Towards Efficient and Secure Authentication: A Novel Protocol with Formal Validation** – Springer ICMLCS 2025 [DOI]

Technical Skills

Languages: Python, Java, JavaScript, C++, C, SQL
Full-Stack Development: MERN Stack (MongoDB, Express.js, React.js, Node.js), HTML, CSS, Bootstrap, FastAPI, Flask, REST API Design
AI / Machine Learning: LangChain, LangGraph, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), FAISS, ChromaDB, Sentence Transformers, Model Context Protocol (MCP), LLM Integration, Prompt Engineering, Scikit-learn, NumPy, Pandas
Backend & Systems: Asynchronous Programming (asyncio), Redis, Task Queues (ARQ, BullMQ), Webhooks, Idempotency, WebSockets (Socket.IO), SQLite, MongoDB, OAuth 2.0, JWT
Engineering Practices: Git, GitHub, Data Structures & Algorithms, Object-Oriented Programming, SDLC, Debugging, Modular Software Architecture, Unit Testing, Structured Logging & Observability

Achievements

- Qualified Stage II, Smart India Hackathon (SIH) 2024.
- Winner, Government IDE Bootcamp 2026 (Edition 3, Phase II) – built CodeSync.ai.